ISSN: 3032-3134 (ONLINE)
InaJBCS Indonesian Journal of Biomedicine and Clinical Sciences

Brief cognitive behavioral therapy for chronic insomnia in a primary care setting: a case report

Dellma Anggini[1], **Mualimin**[2]

[1]Borobudur Community Health Center, [2]Department of Biology Education, Universitas Negeri Yogyakarta

https://doi.org/10.22146/inajbcs.v58i3.28287

Submitted: 2026-01-26
Accepted : 2026-05-04

ABSTRACT

Chronic insomnia with anxiety disorders is a common non-specialist mental health condition frequently encountered in primary care. Although the WHO Mental Health Gap Action Programme (mhGAP) recommends psychosocial interventions at this level, skepticism persists regarding the feasibility of implementing brief psychotherapy within the high-volume workflow of Indonesian community health centers. This case report aims to illustrate the clinical application and operational feasibility of Brief Cognitive Behavioral Therapy (BCBT) in a primary health center setting. We report a 40-year-old female with chronic insomnia accompanied by somatic symptoms, including palpitations and tachycardia, without underlying medical comorbidities. The patient exhibited significant cognitive distortion in the form of sleep state misperception. As the primary treatment modality, a four-session BCBT intervention was conducted every two weeks (30–45 minutes per session). The curriculum included autonomic, behavioral, and cognitive (ABC) model psychoeducation, cognitive restructuring, scheduled worry period, behavioral activation, relaxation, lifestyle modification, and relapse prevention. Evaluation was performed using the insomnia severity index (ISI), subjective units of distress scale (SUDS), and collaborative patient-reported outcome measures (PROM) encompassing three domains: sleep duration, frequency, and anxiety levels. The collaborative goal setting (CGS) of at least 5 hours of sleep, 5 days a week without anxiety, was fully achieved by the third session and remained stable until thereafter. This case illustrates that structured psychotherapy such as BCBT is potentially feasible within a busy PHC workflow, provided there is institutional commitment from facility leadership, a dedicated psychotherapy schedule for trained mental health providers at least once a week, and functional early detection within general outpatient services. While these results are promising for this individual case, further studies are needed to evaluate the scalability and standardized implementation of this model across diverse primary care settings in Indonesia.

ABSTRAK

Insomnia kronis dengan gangguan cemas merupakan kondisi nonspesialis yang sering ditemukan di layanan primer. Meski WHO *Mental Health Gap Action Programme* (mhGAP) merekomendasikan intervensi psikososial di layanan primer, masih terdapat skeptisisme mengenai kelayakan penerapan psikoterapi singkat dalam alur pusat Kesehatan masyarakat yang padat. Laporan kasus ini bertujuan untuk mengilustrasikan aplikasi klinis dan kelayakan operasional Brief Cognitive Behavioral Therapy (BCBT) di Puskesmas. Kami melaporkan seorang perempuan 40 tahun dengan insomnia kronis yang disertai gejala somatik berupa palpitasi dan takikardia tanpa komorbiditas medis yang mendasari. Pasien memiliki distorsi kognitif berupa mispersepsi tidur yang berat (*sleep state misperception*). Sebagai pilihan terapi utama, dilakukan intervensi BCBT sebanyak empat sesi setiap 2 minggu 30-45 menit dengan kurikulum psikoedukasi model *autonomic, behavioral, and cognitive* (ABC), restrukturisasi kognitif, jadwal khawatir, aktivasi perilaku, relaksasi dan modifikasi gaya hidup, serta mencegah kekambuhan. Evaluasi menggunakan *insomnia severity index* (ISI), *subjective units of distress scale* (SUDS), dan *Patient-Reported Outcome Measures* (PROM) kolaboratif yang mencakup tiga domain berupa durasi tidur, frekuensi, dan tingkat kecemasan. *collaborative goal setting* (CGS) berupa tidur setidaknya 5 jam selama 5 hari dalam seminggu tanpa rasa cemas tercapai sepenuhnya pada sesi ketiga dan bertahan stabil. Kasus ini mengilustrasikan bahwa psikoterapi terstruktur seperti BCBT berpotensi layak diterapkan dalam alur kerja Puskesmas yang padat, asalkan terdapat komitmen institusional dari pimpinan fasilitas, adanya jadwal khusus psikoterapi bagi petugas kesehatan jiwa terlatih setidaknya satu kali dalam seminggu, serta berjalannya deteksi dini di pelayanan umum. Meskipun hasil ini menjanjikan untuk kasus individu ini, studi lebih lanjut diperlukan untuk mengevaluasi skalabilitas dan standarisasi implementasi model ini di berbagai pengaturan layanan primer lainnya di Indonesia.

Keywords:
brief cognitive behavioral therapy;
case report;
chronic insomnia;
primary health care;
somatic anxiety

*corresponding author: dellmaa@yahoo.com

INTRODUCTION

Chronic insomnia is a common complaint in primary health care (PHC) settings, with a global prevalence ranging from 10% to 30%, and it often manifests with somatic symptoms that impair daily functioning and productivity.[1,2] Chronic insomnia is also associated with substantial impairment in health-related quality of life (HRQoL) and social functioning.[1,3] From a pathophysiological perspective, insomnia can be understood through the interaction of predisposing, precipitating, and perpetuating factors, as described in Spielman's three-factor model, with cognitive and physiological hyperarousal contributing to the persistence of symptoms.[4,5] Current guidelines recommend behavioral and psychosocial interventions as first-line treatment for chronic insomnia, and the World Health Organization (WHO), through the mhGAP Intervention Guide, also supports the delivery of psychosocial interventions in non-specialized settings.[3,6,7] Nevertheless, a substantial implementation gap remains, particularly in low- and middle-income countries, where the burden of untreated mental disorders at the primary care level remains considerable.[8]

The community health centers (Puskesmas), as the frontline of the Indonesian healthcare system, are mandated to conduct early detection and provide essential mental health care. However, the ideal referral pathway to specialized psychiatric services is often hindered by multiple barriers, including stigma, limited access, financing constraints, and geographical obstacles.[9,10] In this context, the integration of brief psychotherapeutic interventions into primary care may help reduce the treatment gap. Behavioral and psychological interventions for insomnia have demonstrated clinical utility,[11] and brief CBT for insomnia has also shown feasibility when delivered in primary care settings.[12] However, evidence regarding its implementation by primary care physicians in high-volume Indonesian PHC settings remains limited. In addition, the acceptability of cognitive behavioral therapy in the Indonesian community health care has only recently begun to be documented.[13]

This case report is important because it describes a rare but clinically relevant example of physician-led brief cognitive behavioral therapy (BCBT) delivered within a community health center setting. Beyond its clinical relevance, this case provides a practical illustration of how a structured psychological intervention may be adapted into routine primary care workflows in a resource-limited environment. Through the "Mental Health Clinic" initiative, this report describes the clinical application and operational feasibility of BCBT for a 40-year-old woman with chronic insomnia and somatic anxiety, while maintaining the continuity of general medical service flow. This case may provide a practical, context-specific reference for similar primary care settings in Indonesia.

CASE

A 40-year-old woman who worked as a trader and homemaker attended the community health center because she had experienced persistent insomnia for approximately five months. Her sleep problems had worsened in conjunction with marked anxiety, palpitations, and thanatophobia. She also reported a longer history of recurrent somatic complaints, particularly epigastric discomfort and tension-type headaches. During several previous primary care visits, these symptoms had been managed with symptomatic medication. Although the treatment provided temporary relief, the psychological distress underlying the complaints had not been explored. She reported no previous psychiatric diagnosis, chronic medical illness, or long-term medication use. She also denied substance misuse, including dependence on caffeine or tobacco. No family history of mental health disorders or hereditary cardiovascular conditions was reported.

A psychosocial assessment using the home, education/employment, activities, drugs, sexuality, and suicide/depression (HEADSS) framework identified family support as an important protective factor. However, the relocation of her marketplace was associated with reduced occupational functioning and psychomotor agitation.[14] At baseline, the insomnia severity index (ISI) score was 24, which was consistent with severe clinical insomnia.[15] This score was subsequently used to monitor changes throughout the BCBT sessions. Physical examination and electrocardiography showed sinus tachycardia, with a heart rate of 108 beats per minute, without evidence of primary organic pathology. Mental status examination identified sleep-state misperception and cognitive hyperarousal as prominent cognitive features.

The patient underwent BCBT as the primary intervention. The protocol was adapted from brief cognitive-behavioral therapy principles and incorporated ABC model psychoeducation, cognitive restructuring, scheduled worry period, behavioral activation, relaxation, lifestyle modification, and relapse prevention.[16,17] To ensure therapeutic continuity within the community health center's mental health service, each patient was managed consistently by a single designated mental health provider throughout the treatment course. In this case, the intervention was conducted independently by a general practitioner. The protocol was initially planned as a six-session framework, with each biweekly encounter lasting 30–45 minutes.[16]

Every session followed a standardized sequence to ensure clinical consistency and monitor progress from the patient's perspective. Each session began with evaluation using the ISI, the SUDS, and collaborative PROM based on the previously established CGS using a Likert scale of 0-6.[15,16] This was followed by the specific therapeutic intervention and concluded with homework to be practiced between sessions.

The intervention followed a modified BCBT protocol, adapted to align with the standard operating procedures (SOP) for managing mild-to-moderate depression and anxiety at the community health center. Although the institutional SOP outlines a six-session framework, the intervention was completed in four concentrated sessions. This adjustment was based on the patient's early symptomatic improvement and the fact that she achieved her self-defined therapeutic goals sooner than anticipated. By the fourth session, substantial recovery had been observed. The components originally planned for sessions five and six were therefore incorporated into the final encounter to complete the intervention efficiently. The clinical course and progress across sessions are summarized in TABLE 1.

TABLE 1. Summary of clinical progress and therapeutic evaluation

Session	Intervention focus	Anxiety intensity (SUDS 0-100%)	ISI Score	Evaluation
1	Assessment, Safety Planning, & Psychoeducation	100%	24 (severe)	Initiated Amitriptyline & Diazepam. Patient highly anxious (HR: 108 bpm).
2	Cognitive Restructuring	85% to 45%	16 (moderate)	Pharmacotherapy completely discontinued (patient refused due to fear of addiction). Snoring video evidence debunked misperception.
3	Scheduled Worry Period	20%	11 (mild)	No medication; patient able to manage anxiety using only cognitive techniques.
4	Behavioral Activation	10%	6 (normal)	Occupational functioning restored. Brief discussion on relaxation and relapse prevention.
5	Relaxation & Lifestyle Modification	-	-	(Merged into Session 4).
6	Relapse Prevention	-	-	(Merged into Session 4; termination).

During the first session, the patient underwent assessment and received psychoeducation using the ABC model. A collaborative patient-reported outcome measure (PROM) was then established to guide treatment. The PROM included three domains: sleep duration, frequency of adequate sleep per week, and subjective anxiety level.16 The practitioner initially suggested a target of six hours of sleep on five days each week. However, the patient considered sleeping for at least five hours on five days per week without anxiety to be both realistic and beneficial. This agreed target was reviewed at each session. Amitriptyline and diazepam were initially prescribed as adjunctive treatment. However, the patient discontinued both medications before the second session because of her fear of medication dependence. Subsequent treatment therefore focused on psychotherapeutic strategies.

The second session focused on identifying the automatic thoughts associated with the patient's autonomic symptoms. The clinician and patient completed a cognitive restructuring form together, as presented in TABLE 2. During this session, the patient identified her emotions and thoughts, while the counselor guided the identification of objective evidence to challenge the existing cognitive distortions.[17,18] In this session, the patient's spouse—who was present as a collateral source—voluntarily shared a spontaneous home-recording he had captured. This incidental evidence, showing the patient snoring and physically still for 15 minutes, was utilized as a powerful reality testing tool to address the patient's sleep state misperception.

TABLE 2. Cognitive restructuring form (Session 2) (Collaboratively completed by the patient and counselor)

Situation	Emotions & intensity (0-100%)	Automatic thoughts (belief 0-100%)	Evidence supporting the thought	Evidence against the thought (contradictory facts)	Cognitive revision (alternative thought)	New emotions & intensity (0-100%)
Nighttime approaching bedtime	Anxiety (100%) Fear (90%)	"I cannot sleep at all." (100%)	Pacing around at night and engaging in various activities to fill the time.	Husband reported that I actually slept, even if only for a few minutes. Incidental video evidence (spontaneously recorded by the spouse) showing me snoring while asleep.	"I am actually capable of sleeping." (75%)	Anxiety (45%) Fear (20%)

Socratic questioning was used to help the patient compare her automatic belief with the available evidence. Reviewing the inconsistency between her thoughts and the objective evidence was followed by a reduction in reported anxiety and a change in how she interpreted her sleep experience.[17,18]

Doctor: "You are convinced that you did not sleep at all; however, in this video incidentally recorded by your husband, you appear to be fast asleep and snoring. In your opinion, what is actually happening there?"

Patient: (Silent, observing the video) "That is indeed me, Doctor. But it felt as though I was awake all night."

Doctor: "That is what we call sleep state misperception. Physiologically, your body was actually resting. If this evidence shows that you can indeed sleep, is the fear of 'dying from lack of sleep' still entirely true?"

Patient: "It turns out it isn't true, Doctor. Seeing this video makes me feel much calmer."

At the beginning of the third session, the patient reported less cognitive distress. Her SUDS score had decreased to 20%, and her PROM had improved after two weeks of practicing the initial cognitive strategies at home.[15,16] Given the remaining cognitive hyperarousal, the treatment focus shifted to stimulus control through a scheduled worry period (SWP). This intervention was intended to confine worry to a designated time so that it would be less likely to interfere with sleep.[19]

Her anxiety was first framed as an understandable response to the recent relocation of her marketplace. She was then asked to set aside 15–20 minutes in the afternoon as a designated worry period. Outside this period, she recorded intrusive thoughts in a "worry script" and practiced redirecting her attention to ongoing activities. During the SWP, she reviewed the recorded concerns to distinguish between problems that could be addressed constructively and worries that were not productive. The session ended with homework to continue both the scheduled worry period and the worry-script record.[19]

By the beginning of the fourth session, the SUDS score had decreased further to 10%. The PROM scores recorded at the beginning of Sessions 3 and 4 were both 6/6 (100%), indicating that the therapeutic goals defined by the patient had been achieved. In view of this improvement, the patient requested to conclude therapy. The counselor agreed with this collaborative decision and incorporated the remaining components originally planned for Sessions 5 and 6 into the final session.

Behavioral activation (BA) was the main focus of this session because the patient had previously experienced inactivity and feelings of worthlessness. The session included psychoeducation on the inactivity–depression cycle and the development of a Weekly Activity Schedule. Activities were grouped into necessary activities, such as household chores; important activities, such as market-related tasks; and pleasurable activities, such as social or leisure activities. Using a graded task-assignment approach, the patient selected one to three manageable activities each week, beginning with the simplest tasks and progressing as able. Her mood and sense of mastery were reviewed regularly.

Relaxation techniques and lifestyle measures were also introduced to support the maintenance of improvement. The patient practiced deep-breathing exercises and guided imagery, in which she visualized a place where she felt most comfortable, to reduce nighttime anxiety. Affirmations such as “I am capable of sleep” and “I am under God’s protection” were used to reinforce more helpful cognitions. Lifestyle recommendations included regular physical activity for 30 minutes, five times per week, a balanced diet, and avoidance of tobacco smoke to support sleep.

The final session also included relapse-prevention planning based on the ABC framework. The patient was encouraged to recognize early signs of distress, including autonomic arousal and catastrophic thoughts, and to apply the strategies learned during treatment independently. She was also advised to seek professional support if self-management strategies were insufficient during future stressors.

After completing the four-session BCBT protocol, the patient was advised to return if symptoms recurred. During later visits to the primary health center for unrelated minor conditions, such as a common cold, informal follow-up assessments were performed. She continued to report stable emotional functioning and improved sleep quality, with no recurrence of severe insomnia or somatic anxiety. Her final ISI score was 6, which was consistent with clinical remission. No adverse events, paradoxical insomnia, or unexpected psychological distress were observed during the intervention or the informal follow-up period.

At the final evaluation, the patient described feeling relieved and more able to manage her symptoms. She appreciated that the sessions were structured, brief, and practical within her daily routine. She stated, “I feel more in control now. The tools I learned have given me a sense of safety I did not have before.” Overall, she expressed high satisfaction with treatment and described a shift toward active self-management.

The prognosis was considered favorable in view of the rapid reduction in symptoms, good engagement with treatment, and use of cognitive-behavioral strategies outside the sessions. Achieving her self-defined therapeutic goals by the third session also suggested a capacity for continued recovery and adaptive coping with future stressors.

DISCUSSION

This case suggests that BCBT may serve as the main therapeutic approach for addressing autonomic hyperarousal in chronic insomnia accompanied by somatic anxiety, including when pharmacotherapy is not continued.[11,12] Within Spielman’s three-factor model, sleep state misperception appeared to be an important perpetuating factor in this patient.[4,18] Her tachycardia and palpitations were not considered primary cardiac disorders but were clinically interpreted as manifestations of thanatophobia, which may have contributed to sympathetic nervous system hyperactivity.

The patient's clinical improvement became evident as treatment shifted to a non-pharmacological approach. Consequently, the intervention proceeded solely through the BCBT framework. The foundation of this outcome was likely established during the initial encounter through the implementation of the ABC model of psychoeducation and the cultivation of a strong therapeutic alliance.[16,17] The clinical impression suggests that establishing a clear understanding of how cognitive patterns influence autonomic physical symptoms provided a "turning point" for the patient.[17] This was quantitatively reflected by an increase in PROM scores from 0 to 3 by the second session, suggesting the impactful role of early psychological stabilization in the overall recovery process.

A significant clinical breakthrough occurred following the second session through an empirical confrontation technique using video recordings as a tool for reality testing (FIGURE 1). The clinical impression suggests that addressing the belief of "total insomnia" helped reduce the cognitive load and hypervigilance.[17,18] This visual trajectory is consistent with an "early responder" phenomenon, where the inverse correlation between declining anxiety intensity and rising PROM scores (reaching 6/6 by Session 3) suggests the impactful role of cognitive restructuring in this specific case. The maintenance of these scores through the final session suggests that the patient had internalized a sense of self-efficacy, which may have helped sustain her recovery.

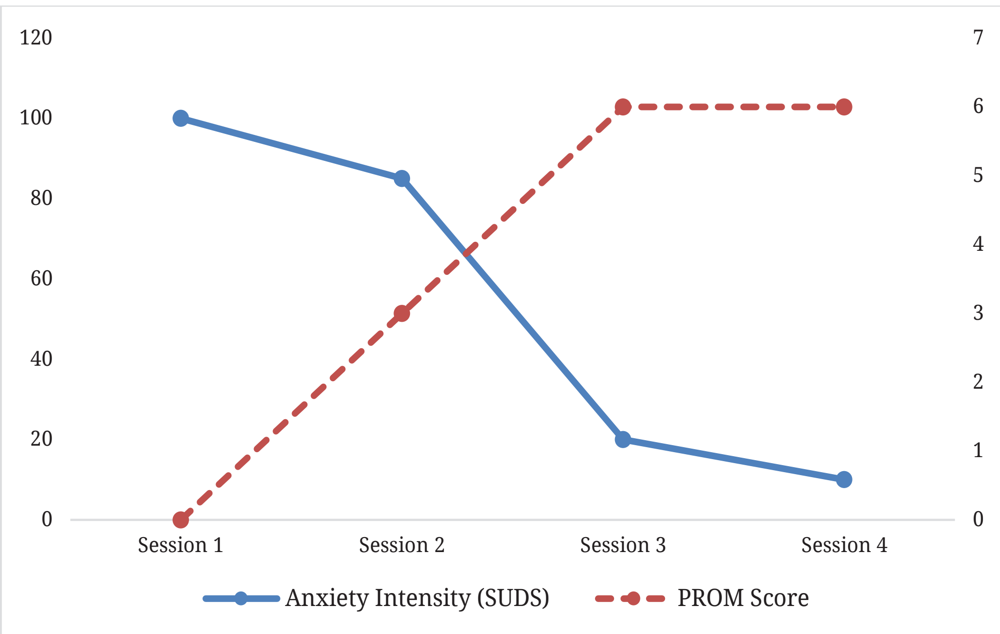

FIGURE 1. Correlation between insomnia and patient-reported outcome.

In addition to the clinical outcome measures, the patient's own account provides insight into the psychological changes associated with BCBT. Her reported relief and empowerment were consistent with improved self-efficacy, which is considered an important mediator in cognitive-behavioral interventions.[16,17] She also valued the brief and structured format of the sessions, describing it as practical within the demands of her daily life. Her statement, *'I feel more in control now'* reflected her ability to use the therapeutic strategies actively rather than remain a passive recipient of care. This increased level of patient involvement contributed to her immediate satisfaction and may also support longer-term recovery and resilience in responding to future stressors.

Meaningful clinical improvement was observed even though the intervention followed a general BCBT protocol for anxiety and depression rather than a specialized Cognitive Behavioral Therapy for Insomnia (CBT-I) framework. CBT-I commonly includes structured components such as stimulus control, sleep restriction therapy, and sleep hygiene.[3,7,11,20] In this patient, addressing somatic anxiety and cognitive distortions appeared sufficient to relieve the main complaint of insomnia, suggesting a possible transdiagnostic benefit.[16,17] The emphasis on scheduled worry periods and relaxation techniques, instead of rigid sleep scheduling, was associated with reduced physiological hyperarousal.[16,19] This approach may be relevant in primary care when sleep disturbance is closely related to anxiety.[12,16]

This case also demonstrates how clinical adjustment may improve treatment efficiency in a high-volume community health center. Although the institutional SOP for mild-to-moderate anxiety and depression generally recommends six sessions, the intervention was condensed into four intensive sessions. This decision was guided by the patient's rapid improvement, early achievement of her therapeutic goals, and final Insomnia Severity Index (ISI) score of 6. Relaxation and relapse-prevention components were incorporated into the final session to make efficient use of the available consultation time without compromising the perceived integrity of treatment. This case suggests that primary care physicians and other dedicated mental health workers may be able to provide holistic mental health care in resource-limited settings when structured time management is combined with a patient-centered approach.

The successful delivery of this brief intervention requires a client-centered approach from the practitioner. In this primary health center, the SOP is implemented by a multidisciplinary team consisting of general practitioners, mental health nurses, and other dedicated mental health workers. One challenge observed in this local setting is the possibility that practitioners may unintentionally introduce personal values or moral judgments into the therapeutic process. Moving away from a neutral and empathetic stance may reduce the usefulness of the protocol and interfere with the development of a strong therapeutic alliance. For BCBT to be implemented effectively within this institutional setting, practitioners need to maintain therapeutic neutrality and consistently apply a patient-centered model. The intervention should remain collaborative rather than directive or judgmental to support consistent outcomes across providers.

The sustainability of this intervention within the local public health center appeared to depend largely on organized system-level support. This case suggests that brief psychotherapy may be incorporated into routine primary care when there is strong institutional commitment from the Head of the community health center.[6,12,13] Important operational measures include allocating at least one day each week for mental health consultations by a trained provider and ensuring that early mental health screening is implemented in general outpatient services. Such support can help establish mental health care as a standard component of primary health services rather than an isolated activity.

Complex anxiety disorders are often managed at the secondary-care level. However, this case illustrates how a primary care physician may help narrow the treatment gap by delivering evidence-based psychological interventions within the community.6,12 This approach is consistent with the WHO mhGAP framework, which supports the decentralization of mental health services to improve access and reduce stigma.

Several limitations of this case report should be considered. First, the clinical evaluation did not include widely validated screening instruments for anxiety or depression, such as the PHQ-9 or GAD-7. Instead, the patient's progress was monitored using the SUDS and PROM. Although these measures captured the patient's own perception of her symptoms and daily functioning, future studies should include standardized assessment tools to obtain more objective data that can be compared across settings.

A second limitation relates to the use of the community health center's general SOP framework. This SOP is applied uniformly to cases of mild-to-moderate anxiety or depression, and there are currently no diagnosis-specific protocols that require particular psychotherapeutic modalities. Therefore, the outcome of this intervention may have been influenced by the patient's individual characteristics and the therapeutic alliance established during treatment. Further studies involving a broader population are needed to evaluate the clinical usefulness and scalability of this condensed BCBT protocol in primary care settings.

CONCLUSION

In this case, the BCBT intervention was associated with substantial clinical improvement and a return to the patient's premorbid level of functioning. The patient described the approach as practical and empowering, as it helped her move from passive symptom management toward more active self-management and confidence in using the strategies she had learned. The intervention also appeared workable within a busy primary health care setting. Its implementation, however, requires institutional support, a dedicated psychotherapy schedule for a trained mental health provider at least once each week, and an effective early-detection process within general outpatient services. As these findings are based on a single case, further studies are needed to examine the scalability and adaptability of this model across different primary care settings in Indonesia.

ACKNOWLEDGMENT

The client provided signed informed consent for publication. All identifiable details were altered to ensure privacy. The authors declare no conflict of interest. Sincere gratitude is extended to the patient and her family for their invaluable collaboration. Special thanks are also due to the Head of Borobudur Community Health Center for the administrative support and permission provided for this study.

REFERENCES

1. Morin CM, Jarrin DC. Epidemiology of insomnia: prevalence, course, risk factors, and public health burden.

Sleep Med Clin 2022; 17(2):173-191. https://doi.org/10.1016/j.jsmc.2022.03.003
2. Bhaskar S, Hemavathy D, Prasad S. Prevalence of chronic insomnia in adult patients and its correlation with medical comorbidities. J Family Med Prim Care 2016; 5(4):780-4. https://doi.org/10.4103/2249-4863.201153
3. Riemann D, Baglioni C, Bassetti C, Bjorvatn B, Dolenc Groselj L, Ellis JG, *et al*. European guideline for the diagnosis and treatment of insomnia. J Sleep Res 2017; 26(6):675-700. https://doi.org/10.1111/jsr.12594
4. Spielman AJ, Caruso LS, Glovinsky PB. A behavioral perspective on insomnia treatment. Psychiatr Clin North Am 1987; 10(4):541-53. https://doi.org/10.1016/S0193-953X(18)30532-X
5. Levenson JC, Kay DB, Buysse DJ. The pathophysiology of insomnia. Chest 2015; 147(4):1179-92. https://doi.org/10.1378/chest.14-1617
6. World Health Organization. mhGAP intervention guide for mental, neurological and substance use disorders in non-specialized health settings: Mental Health Gap Action Programme (mhGAP). Geneva: World Health Organization; 2010.
7. Qaseem A, Kansagara D, Forciea MA, Cooke M, Denberg TD. Management of chronic insomnia disorder in adults: a clinical practice guideline from the American college of physicians. Ann Intern Med 2016; 165(2):125-33. https://doi.org/10.7326/M15-2175
8. Thornicroft G, Chatterji S, Evans-Lacko S, Gruber M, Sampson N, Aguilar-Gaxiola S, *et al.* Undertreatment of people with major depressive disorder in 21 countries. Br J Psychiatry 2017; 210(2):119-24. https://doi.org/10.1192/bjp.bp.116.188078
9. Hartini N, Fardana NA, Ariana AD, Wardana ND. Stigma toward people with mental health problems in Indonesia. Psychol Res Behav Manag 2018; 11:535-41. https://doi.org/10.2147/PRBM.S175251
10. Munira L, Liamputtong P, Viwattanakulvanid P. Barriers and facilitators to access mental health services among people with mental disorders in Indonesia: a qualitative study. Belitung Nurs J 2023; 9(2):110-17. https://doi.org/10.33546/bnj.2521
11. Edinger JD, Arnedt JT, Bertisch SM, Carney CE, Harrington JJ, Lichstein KL, *et al.* Behavioral and psychological treatments for chronic insomnia disorder in adults: an American Academy of Sleep Medicine clinical practice guideline. J Clin Sleep Med 2021; 17(2):255-62. https://doi.org/10.5664/jcsm.8986
12. Pigeon WR, Funderburk JS, Cross W, Bishop TM, Crean HF. Brief CBT for insomnia delivered in primary care to patients endorsing suicidal ideation: a proof-of-concept randomized clinical trial. Transl Behav Med 2019; 9(6):1169-77. https://doi.org/10.1093/tbm/ibz108
13. Bouman TK, Lommen MJJ, Setiyawati D. The acceptability of cognitive behaviour therapy in Indonesian community health care. Cogn Behav Ther 2022; 15:e26. https://doi.org/10.1017/S1754470X22000228
14. Goldenring JM, Rosen DS. Getting into adolescent heads: an essential update. Contemp Pediatr 2004; 21(64):64-90.
15. Bastien CH, Vallières A, Morin CM. Validation of the Insomnia Severity Index as an outcome measure for insomnia research. Sleep Med 2001; 2(4):297-307. https://doi.org/10.1016/S1389-9457(00)00065-4
16. Cully JA, Teten AL. A therapist's guide to brief cognitive behavioral therapy. Houston: Department of Veterans Affairs, South Central MIRECC; 2008.
17. Clark DA, Beck AT. Cognitive therapy

of anxiety disorders: science and practice. New York: Guilford Press; 2010.

18. Joo EH, Altier HR, Selai C, Gratton MK, Kim-Dahl A, Allen H, *et al*. Neurobiological mechanisms of sleep state misperception in insomnia disorder: a theoretical review. Sleep Med Rev 2025; 81:102096. https://doi.org/10.1016/j.smrv.2025.102096
19. McGowan SK, Behar E. A preliminary investigation of stimulus control training for worry: effects on anxiety and insomnia. Behav Modif 2013; 37(1):90-112. https://doi.org/10.1177/0145445512455661
20. Octaviani C, Prasetya H, Murti B. Effectiveness of cognitive behavioral therapy for insomnia on sleep quality in patients with insomnia: a meta-analysis. Indones J Med 2023; 8(3):289-301. https://doi.org/10.26911/theijmed.2023.8.3.652